

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2–Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2–Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	DHIVYADHARSAN M.S	Reg. No.:	192511465
Section/Batch:	2025	Date:	24/07/2026

Code of Conduct

I, DHIVYADHARSAN M.S (Name/RegNo)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: DHIVYADHARSAN M.S.

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answers using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- None negative marking. Unanswered questions carry zero marks.

Section/Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1- 5	25
LinkedList Data Structure	List Operations	6–10	25
Queue Data Structures	Queue Operations	11- 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16- 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

 Binary Search
5 marks

Q2

 Stack
5 marks

Q3

 Stack
5 marks

Q4

 Stack
5 marks

Q5

 Stack - Balancing Symbols
5 marks

Q6

 Singly Linked List
5 marks

Q7

 Stack
5 marks

Q8

 Linked List
5 marks

Q9

 Linked List
5 marks

Q10

 Queue
5 marks


10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low < high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int n, int key)
4 {
5     int low = 0, high = n - 1;
6
7     while (low <= high)
8     {
9         int mid = low + (high - low) / 2;
10
11         if (arr[mid] == key)
12             return mid;
13         else if (arr[mid] < key)
14             low = mid + 1;
15         else
16             high = mid - 1;
17     }
18
19     return -1;
20 }
21
22 int main()
23 {
24     int n, key;
25     scanf("%d", &n);
26
27     int arr[n];
28     for(int i = 0; i < n; i++)
29         scanf("%d", &arr[i]);
30
31     scanf("%d", &key);
32
33     printf("%d", binarySearch(arr, n, key));
34
35     return 0;
36 }
```

Test Input

```
5
1 3 5 7 9
7
```

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop()  
2 {  
3     if(top == -1)  
4     {  
5         printf("Underflow");  
6         return -1;  
7     }  
8  
9     return stack[top--];  
10 }
```

Test Input

stdin input (optional)

Output

⏱ Pending manual review

✔ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

 Binary Search
5 marks

Q2

 Stack
5 marks

Q3

 Stack
5 marks

Q4

 Stack
5 marks

Q5

 Stack - Balancing Symbols
5 marks

Q6

 Singly Linked List
5 marks

Q7

 Stack
5 marks

Q8

 Linked List
5 marks

Q9

 Linked List
5 marks

Q10

 Queue
5 marks


10/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
  top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int stack[5];
2 int top;
3
4 void init()
5 {
6     top = -1;
7 }
```

Test Input

stdin input (optional)

Output

 Pending manual review

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QUESTIONS

Q1

Binary Search

5 marks



Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8

Linked List

5 marks



Q9

Linked List

5 marks



Q10

Queue

5 marks



10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {  
    while(stack[top] != '(') {  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if(ch == '(') {  
2 {  
3     while(stack[top] != '(') {  
4         {  
5             postfix[j++] = pop();  
6         }  
7         pop();  
8 }
```

Test Input

Output

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
    pop();
} else {
    return 0;
}
```

✂ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if((ch == ']' && stack[top] == '[') ||
2   (ch == '}' && stack[top] == '{') ||
3   (ch == '[' && stack[top] == '['))
4 {
5     pop();
6 }
7 else
8 {
9     return 0;
10 }
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp = head;
2
3 while(temp != NULL)
4 {
5     print(temp->data);
6     temp = temp->next;
7 }
```

Test Input

stdin input (optional)

Output

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 POP(stack)
2 {
3     if(top == -1)
4         printf("Underflow");
5     else
6         return stack[top--];
7 }
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks



Q2
Stack
5 marks



Q3
Stack
5 marks



Q4
Stack
5 marks



Q5
Stack - Balancing Symbols
5 marks



Q6
Singly Linked List
5 marks



Q7
Stack
5 marks



Q8
Linked List
5 marks



Q9
Linked List
5 marks



Q10
Queue
5 marks



10/10 answered

Q8 Linked List

5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while(current != NULL)
2 {
3     next = current.next;
4     current.next = prev;
5     prev = current;
6     current = next;
7 }
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks



Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8

Linked List

5 marks



Q9

Linked List

5 marks



Q10

Queue

5 marks



10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head
```

```
head = new
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new->next = head;
2 head = new;
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE()
2 {
3     if(front == -1)
4     {
5         printf("Underflow");
6         return -1;
7     }
8     else
9     {
10        return Q[front++];
11    }
12 }
```

Test Input

stdin input (optional)

Output

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✔ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 ✓
Stack
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Stack
5 marks

Q4 ✓
Stack
5 marks

Q5 ✓
Doubly Linked List
5 marks

Q6 ✓
Queue
5 marks

Q7 ✓
Queue - Deque
5 marks

Q8 ✓
Binary Search
5 marks

Q9 ✓
Circular Queue
5 marks

Q10 ✓
Stack - Postfix Expression
5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 PEEK(stack)
2 {
3     return stack[top];
4 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192511465RA_6a702a21d316e6_38770861\main.c:1:1:
warning: return type defaults to 'int' [-Wimplicit-int]
  1 | PEEK(stack)
    | ^~~~
C:\Windows\TEMP\sandbox_192511465RA_6a702a21d316e6_38770861\main.c: In function 'main':
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks


10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if ((rear + 1) % MAX == front)
2 {
3     printf("Queue Full");
4 }
5 else
6 {
7     rear = (rear + 1) % MAX;
8     queue[rear] = vehicle;
9 }
```

Test Input

stdin input (optional)

Output

🕒 Pending manual review

✅ Submitted



CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack

5 marks

**Q2**

Circular Queue

5 marks

**Q3**

Stack

5 marks

**Q4**

Stack

5 marks

**Q5**

Doubly Linked List

5 marks

**Q6**

Queue

5 marks

**Q7**

Queue - Deque

5 marks

**Q8**

Binary Search

5 marks

**Q9**

Circular Queue

5 marks

**Q10**

Stack - Postfix Expression

5 marks



10/10 answered

Q3 Stack

5 marks

Point out the error.

if top == MAX:

Overflow

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (top == MAX - 1)
2 {
3     printf("Overflow");
4 }
```

Test Input

stdin input (optional)

Output

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks


10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 POP()
2 {
3     if (top == -1)
4         printf("Underflow");
5 }
```

Test Input

stdin input (optional)

Output

🕒 Pending manual review

✅ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack

5 marks


Q2

Circular Queue

5 marks


Q3

Stack

5 marks


Q4

Stack

5 marks


Q5

Doubly Linked List

5 marks


Q6

Queue

5 marks


Q7

Queue - Deque

5 marks


Q8

Binary Search

5 marks


Q9

Circular Queue

5 marks


Q10

Stack - Postfix Expression

5 marks



10/10 answered

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deleteNode(Node* del)
2 {
3     if (del == NULL)
4         return;
5
6     if (del->prev != NULL)
7         del->prev->next = del->next;
8
9     if (del->next != NULL)
10        del->next->prev = del->prev;
11
12    free(del);
13 }
```

Test Input

stdin input (optional)

Output

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Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks


10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data)
2 {
3     if (rear == MAX - 1)
4         return;
5
6     queue[++rear] = data;
7 }
```

Test Input

Output

 Pending manual review

 Submitted



CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack

5 marks


Q2

Circular Queue

5 marks


Q3

Stack

5 marks


Q4

Stack

5 marks


Q5

Doubly Linked List

5 marks


Q6

Queue

5 marks


Q7

Queue - Deque

5 marks


Q8

Binary Search

5 marks


Q9

Circular Queue

5 marks


Q10

Stack - Postfix Expression

5 marks



10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {
    if (front == 0) front = MAX - 1;
    else front = front - 1;
    deque[front] = data;
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data) {
2     if (front == 0)
3         front = MAX - 1;
4     else if (front > 0)
5         front = front - 1;
6
7     deque[front] = data;
8 }
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks


10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int l, int r, int x) {
2     int mid = l + (r - l) / 2;
3
4     if (arr[mid] == x)
5         return mid;
6
7     if (arr[mid] > x)
8         return binarySearch(arr, l, mid - 1, x);
9
10    return binarySearch(arr, mid + 1, r, x);
11 }
```

Test Input

Output

 Pending manual review

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 ✓
Stack
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Stack
5 marks

Q4 ✓
Stack
5 marks

Q5 ✓
Doubly Linked List
5 marks

Q6 ✓
Queue
5 marks

Q7 ✓
Queue - Deque
5 marks

Q8 ✓
Binary Search
5 marks

Q9 ✓
Circular Queue
5 marks

Q10 ✓
Stack - Postfix Expression
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    front = (front + 1) % MAX;
    return data;
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue() {
2     if (front == -1) return -1;
3
4     int data = queue[front];
5
6     if (front == rear)
7         front = rear = -1;
8     else
9         front = (front + 1) % MAX;
10
11     return data;
12 }
```

Test Input

stdin input (optional)

Output

⌚ Pending manual review

✓ Submitted

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks


10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
    if (isdigit(expression[i])) push(expression[i] - '0');
    else {
        int op1 = pop();
        int op2 = pop();
        // perform operation: result = op1 + op2;
        push(result);
    }
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while (expression[i] != '\0') {
2     if (isdigit(expression[i]))
3         push(expression[i] - '0');
4     else {
5         int op2 = pop();
6         int op1 = pop();
7
8         push(result);
9     }
10 }
```

Test Input

Output

 Pending manual review

 Submitted